Functional Annotation Report: Mouse *Csn3* (Kappa-casein, UniProt P06796)

Gene/Protein Identity Verification

Before presenting findings, the target identity was confirmed against the provided UniProt record and the published literature:

Attribute	Target (UniProt P06796)	Literature match
Gene symbol	*Csn3* (synonyms *Csn10*, *Csnk*)	✓ *Csnk* used in the mouse knockout paper (P 16698927)
Protein	Kappa-casein, Precursor	✓ Consistent across all cited milk-protein literature
Organism	*Mus musculus* (mouse)	✓ Knockout and expression data are from mouse
Family	Kappa-casein family	✓ Member of secretory calcium-binding phosphoprotein (SCPP) family (P 21245413)
Domain	Casein_kappa (IPR000117 / PF00997)	✓ Consistent with chymosin-cleavable micelle-stabilizing architecture

Verification outcome: The gene symbol *Csn3* unambiguously matches the kappa-casein protein described in UniProt P06796. There is no cross-species or paralog ambiguity in the functional literature. Note: the unrelated "casein kinase" genes *Csnk1/Csnk2* are distinct serine/threonine kinases and are **NOT** this gene; they were explicitly excluded. This report therefore proceeds with full confidence in the target identity.

Summary

Mouse *Csn3* encodes kappa-casein, a small secreted milk protein whose primary biological function is the colloidal stabilization of the casein micelle. Kappa-casein is not an enzyme, transporter, or signaling receptor; it is a structural/surfactant protein. It is the *only* calcium-insensitive member of the four-casein family, and this property is the crux of its function: the calcium-sensitive alpha-s1, alpha-s2, and beta caseins would precipitate in the high calcium/phosphate environment of milk, but kappa-casein coats the calcium-phosphate/casein assemblies and projects a hydrophilic, charged "hairy brush" that sterically and electrostatically holds the entire colloid in soluble suspension (PMID 12778565; PMID 21245413).

Kappa-casein carries out its function **extracellularly**, in secreted milk, although micelle assembly begins intracellularly in the Golgi vesicles of lactating mammary epithelial cells, where 10–20 nm casein particles first form around calcium-phosphate nanoclusters (PMID 21112300). Its essential, non-redundant role was established genetically: kappa-casein-null (*Csnk*$^{-/-}$) female mice fail to lactate because casein micelles are destabilized in the mammary lumen, even though expression of all other milk proteins is unaffected (PMID 16698927).

Beyond its structural role, kappa-casein is the defining physiological substrate of the aspartic protease **chymosin (rennin)**: cleavage of a single Phe–Met peptide bond splits the protein into a membrane-anchored N-terminal para-kappa-casein and a soluble C-terminal (glyco)macropeptide, stripping away the stabilizing brush and triggering milk coagulation (PMID 42440191; PMID 12778565). The released glycomacropeptide is itself a bioactive, prebiotic, immunomodulatory peptide. Kappa-casein additionally behaves as an ATP-independent "holdase" molecular chaperone and, as an intrinsically disordered protein, can convert to amyloid fibrils under destabilizing conditions.

Key Findings

1. Primary function: steric/colloidal stabilization of the casein micelle

The core function of kappa-casein is to stabilize casein micelles — the colloidal particles that transport calcium, phosphate, and protein to the neonate. Kappa-casein forms a "stabilizing hairy brush" on the surface of the micelle and is the only calcium-insensitive casein; it prevents the calcium-sensitive alpha-s and beta caseins from precipitating in the presence of

milk's high calcium and phosphate concentrations (PMID 12778565). The abstract states directly: *"Casein micelles in milk are stable colloidal particles with a stabilizing hairy brush of kappa-casein."* When this surface layer is removed or cleaved, the micelle is destabilized and aggregates. Evolutionary analysis reinforces the point: *"Ca-insensitive casein (κ-casein) does not usually interact with Ca but instead stabilizes the micelle"* (PMID 21245413). The native physiological role is unambiguous — a recent biophysical study notes that kappa-casein *"stabilizes casein micelles in milk but can also form amyloid fibrils under destabilizing conditions"* (PMID 41865247).

2. Physiological substrate of chymosin: the Phe–Met scissile bond and milk coagulation

Kappa-casein is the specific physiological substrate of the aspartic protease chymosin (rennin). Chymosin *"specifically cleaves the Phe105-Met106 bond in κ-casein to initiate milk coagulation"* (bovine numbering), making it indispensable for cheese production (PMID 42440191). This single cleavage splits the protein into two fragments: the hydrophobic, membrane-anchored **para-kappa-casein** (N-terminal) and the soluble, hydrophilic **caseinomacropeptide / glycomacropeptide** (C-terminal). The mechanistic consequence is loss of the steric/electrostatic brush: *"rennet cleaves kappa-casein into casein macropeptide and para-kappa-casein, thereby destabilizing the casein micelle and resulting in aggregation and gel formation of the micelles"* (PMID 12778565). In the mouse protein (P06796), the corresponding chymosin cleavage site is annotated at position 118/119.

3. ATP-independent molecular (holdase) chaperone activity

Milk caseins, including kappa-casein, function as intrinsically disordered, ATP-independent molecular chaperones in the same functional class as lens alpha-crystallins. They *"operate in an ATP-independent manner to stabilize proteins and prevent their misfolding and subsequent aggregation into amorphous particles or highly ordered amyloid fibrils"* (PMID 29442498). Casein globules *"act as holdase-type chaperones for unfolded or partially folded proteins, preventing their aggregation"* (PMID 42715524). Notably, kappa-casein is itself both a chaperone and a client/model amyloid-forming protein under destabilizing conditions (PMID 41865247). This chaperone activity is a biochemical extension of the same surface-adsorption/colloidal-sequestration mechanism underlying micelle stabilization.

4. Expression: lactation-specific, mammary epithelial, PTM-dependent

Csn3 sits within the mammary casein gene locus and is co-expressed with the other caseins, alpha-lactalbumin, and whey acidic protein under lactogenic transcriptional control. Single-cell expression work shows *"highly correlated intra- and interchromosomal coexpression of milk proteins (the casein genes, lactalbumin-alpha and whey acidic protein) and one of its transcriptional regulators (Elf5)"* (PMID 18180458). Functional maturation requires post-translational modification — *"caseins and lactoferrin require intracellular phosphorylation"* (PMID 42634117). UniProt annotates the mouse protein as a secreted precursor with a cleaved signal peptide (residues 1–21), phosphorylation at Thr123, Ser162, and Ser178, and mucin-type O-linked (GalNAc) glycosylation at Thr134, Thr144, Thr155, and Ser162 — all concentrated in the C-terminal macropeptide that forms the stabilizing brush (Finding F005).

5. Genetic proof of essentiality: the *Csnk*$^{-/-}$ knockout mouse

The most decisive evidence for kappa-casein's function comes from mouse genetics. Shekar et al. generated a kappa-casein null mouse and found that homozygous null females *"did not suckle their pups and failed to lactate because of destabilization of the micelles in the lumina of the mammary gland"* (PMID 16698927). Crucially, *"the mutant kappa-casein allele did not affect the expression of other milk proteins in Csnk*$^{-/-}$ *females,"* demonstrating that lactation failure is specifically attributable to loss of the micelle-stabilizing function and not to a general collapse of milk-protein synthesis. This is direct, loss-of-function, in vivo proof that kappa-casein is essential and non-redundant for lactation and successful reproduction in the mouse.

6. The glycomacropeptide (CMP/GMP) is a bioactive signaling fragment

The C-terminal fragment released by chymosin is not merely waste — the glycomacropeptide (GMP) *"was shown under in vitro and in vivo conditions to exert a number of activities that regulate the physiology of important body systems, namely the gastrointestinal, endocrine, and immune systems"* (PMID 30870995). Specific, glycan-dependent effects include preservation of colonic mucus and prebiotic enrichment of *Bifidobacterium*: *"Supplementation of highly sialylated CGMP not only prevented WSD-induced mucus defects but also altered the gut microbiota composition, enhancing beneficial bacterial genera, particularly Bifidobacterium"* (PMID 41370273). GMP also exerts anti-inflammatory effects via increased *Lactobacillus*/*Bifidobacterium*, elevated TGF-β, and inhibited mast-cell activation (PMID 28049206). Thus the molecule has a "second life" as a signaling peptide after proteolytic release.

7. Evolutionary origin: a calcium-insensitive SCPP family member

All casein genes belong to the **secretory calcium-binding phosphoprotein (SCPP)** gene family, which *"arose by gene duplication"* from ancestral tooth/enamel mineralization genes before the origin of mammals (PMID 21245413). Within the family there are two functionally distinct types: calcium-sensitive caseins (alpha-s, beta) that bind calcium but precipitate at high calcium, and the calcium-insensitive kappa-casein that does not usually interact with calcium but instead stabilizes the micelle. Both types are required for a functional micelle — the evolutionary innovation that enabled lactation was the pairing of calcium-sequestering caseins with a calcium-insensitive stabilizer.

8. Micelle architecture and biosynthesis

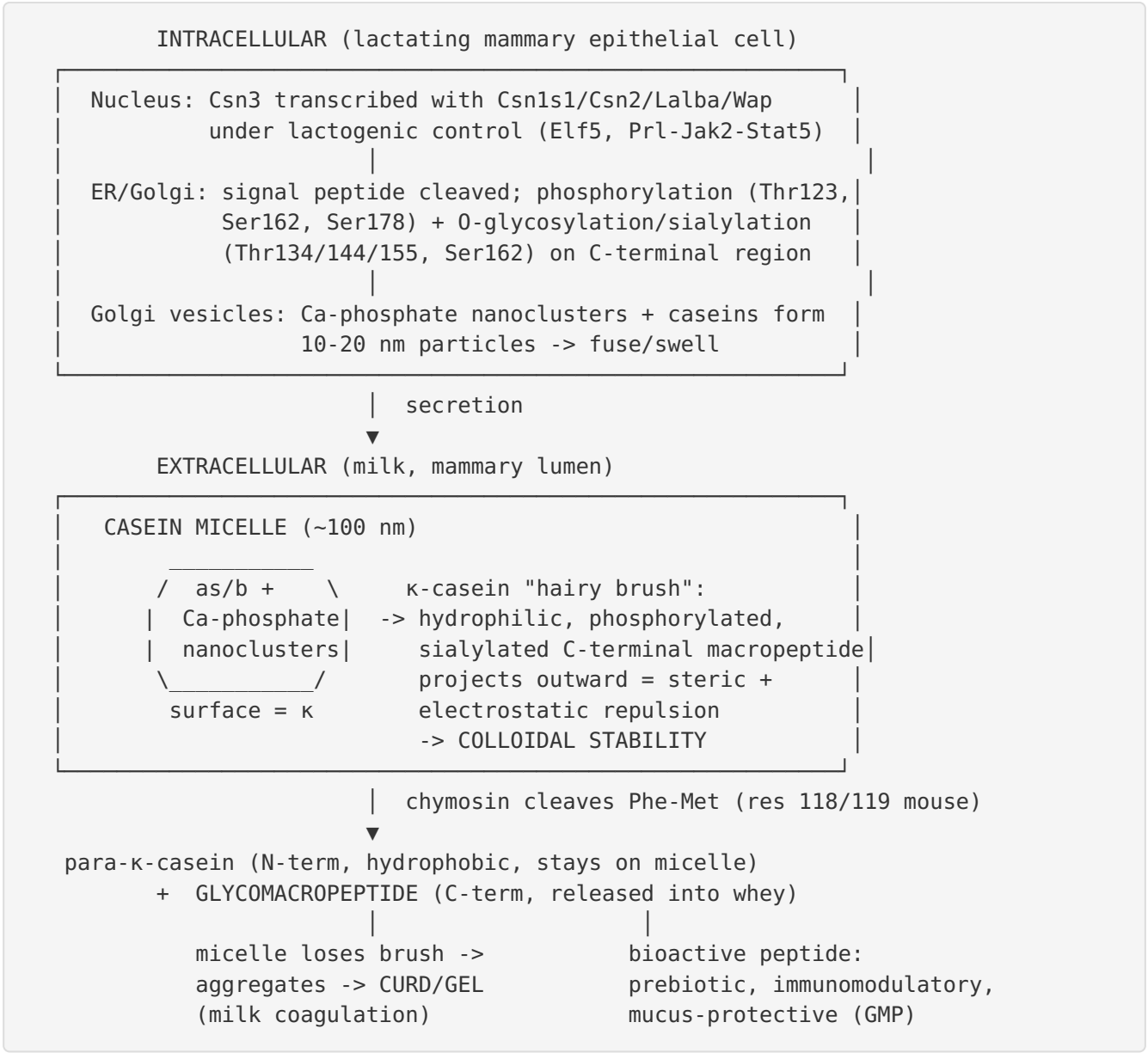
Structural scattering studies (SAXS/SANS) model the casein micelle as a spongelike, hierarchical colloid in which ~2 nm calcium-phosphate nanoclusters anchor the phosphorylated caseins, forming 10–40 nm internal regions connected by water channels into a porous ~100 nm particle (PMID 21112300; PMID 22381008; PMID 25388767). These models are consistent with *"the observation of 10- to 20-nm casein particles in the Golgi vesicles of lactating cells: upon aggregation, those particles would rearrange, fuse, and/or swell to form the spongelike micelle"* (PMID 21112300). Caseins *"self-assemble into nanometer-sized colloids, referred to as 'micelles'"* (PMID 33034319), behaving as "sticky spheres." Kappa-casein locates preferentially at the surface, limiting micelle growth and conferring colloidal stability.

9. Intrinsic disorder underlies all its functions

Kappa-casein is a natively unfolded / intrinsically disordered protein with little stable secondary structure, a feature shared across the casein family (PMID 24162501; PMID 29442498). This conformational flexibility is the physical basis for its amphiphilic surface adsorption at the micelle interface, its holdase chaperone activity, and — under destabilizing conditions — its conversion to amyloid fibrils (PMID 41865247). Proteolytic fragments of kappa-casein are themselves bioactive disordered peptides; for example, the human kappa-casein fragment "lactaptin"/RL2 induces apoptosis selectively in cancer cells, in the context that *"intrinsically disordered proteins play a central role in dynamic regulatory and assembly processes in the cell"* (PMID 31408975).

Mechanistic Model / Interpretation

Kappa-casein can be understood as a **molecular surfactant and growth-terminator** for the casein micelle. The following model synthesizes the findings:



The unifying principle is **amphiphilicity rooted in intrinsic disorder**. Kappa-casein has a hydrophobic N-terminal domain (para-kappa-casein) that associates with the micelle interior/surface and a highly hydrophilic, negatively charged, glycosylated C-terminal domain (macropeptide) that extends into solvent. This "block-copolymer surfactant" architecture:

1. **Caps micelle growth** — kappa-casein on the surface terminates the self-association of the calcium-sensitive caseins, setting the ~100 nm size.
2. **Confers colloidal stability** — the charged, hydrated brush generates steric and electrostatic repulsion between micelles, keeping them dispersed.
3. **Creates a proteolytic "trigger"** — the single exposed Phe–Met bond at the domain junction makes the entire colloidal system switchable: one cleavage collapses stability, which is exploited physiologically (neonatal digestion/clotting) and industrially (cheese).
4. **Releases a signaling cargo** — the cleaved macropeptide acts downstream as a bioactive gut peptide.

This explains why the *Csnk*$^{-/-}$ phenotype is so clean: without the growth-terminator/stabilizer, the remaining caseins aggregate in the lumen, milk cannot be secreted as a stable fluid, and lactation fails — despite normal synthesis of every other milk protein.

Comparison of the casein family

Casein	Gene	Calcium sensitivity	Role in micelle	Position
αs1	*Csn1s1*	Sensitive (precipitates)	Calcium/phosphate sequestration, bulk	Interior
αs2	*Csn1s2*	Sensitive	Calcium/phosphate sequestration	Interior
β	*Csn2*	Sensitive	Calcium/phosphate sequestration, hydrophobic core	Interior
κ	***Csn3***	**Insensitive**	**Surface stabilizer / growth terminator**	**Surface**

Evidence Base

PMID	Title (abbrev.)	Contribution
16698927	*kappa-casein-deficient mice fail to lactate*	**Decisive genetic proof**: in-vivo loss-of-function; micelle destabilization in mammary lumen; lactation failure; other milk proteins unaffected
12778565	*Impaired rennetability of heated milk*	Defines the "stabilizing hairy brush"; describes rennet cleavage products and destabilization/ gelation mechanism
42440191	*Microbial engineering of bovine chymosin*	Specifies the Phe105–Met106 scissile bond and that cleavage initiates coagulation
21245413	*Evolution of milk casein genes from tooth genes*	SCPP evolutionary origin; κ-casein as the calcium-insensitive stabilizer
29442498	*Proteostasis… ATP-independent chaperones: crystallins and caseins*	Establishes caseins as ATP-independent holdase chaperones
42715524	*Amorphous Calcium Phosphate… Casein Microarchitecture*	Casein globules as holdase chaperones
18180458	*Promiscuous gene expression… thymic epithelial cells*	Co-regulated mammary milk-protein expression program (casein genes + Elf5)
42634117	*Rational Design of Precision-Fermented Milk Proteins*	Caseins require intracellular phosphorylation for maturation
30870995	*Glycomacropeptide Bioactivity and Health*	GMP regulates GI/endocrine/immune physiology
41370273	*Casein GMP improves colonic mucus function*	Sialylation-dependent prebiotic/mucosal GMP function
28049206	*GMP on Allergy: microbiota, TGF-β, mast cells*	Immunomodulatory mechanism of GMP
21112300	*How to squeeze a sponge: casein micelles (SAXS)*	Golgi-vesicle biosynthesis; spongelike micelle model
22381008	*Casein micelles and their internal structure*	Nanocluster model; κ-casein limits self-association

PMID	Title (abbrev.)	Contribution
25388767	*Structural heterogeneity... SANS contrast variation*	Core-shell micelle structure
33034319	*Casein micelles in milk as sticky spheres*	Colloidal self-assembly behavior
41865247	*Sorbitol modulates κ-casein amyloid fibrils*	Native stabilizing role vs. amyloidogenic plasticity
24162501	*DMSO-induced conformational changes (ROA)*	Caseins are natively disordered
31408975	*Human κ-casein fragment with antitumor properties*	Intrinsic disorder; bioactive proteolytic fragment

Consistency of the evidence base: The findings are mutually reinforcing across independent lines — genetics (knockout), biophysics (scattering/amyloid), biochemistry (chymosin kinetics, chaperone assays), evolution (SCPP phylogeny), and physiology (GMP bioactivity). No cited study contradicts the central claim that kappa-casein's primary role is colloidal micelle stabilization. The knockout (P 16698927) is the single highest-quality, most precise piece of evidence because it is a targeted, in-vivo, loss-of-function experiment with a specificity control (other milk proteins normal).

Limitations and Knowledge Gaps

1. **Bovine-centric mechanistic detail.** Much of the molecular detail (the Phe105–Met106 scissile bond, the "hairy brush," SAXS/SANS micelle models) derives from *bovine* kappa-casein and bovine milk. Mouse-specific biophysical data are sparser; the mouse chymosin cleavage site (res 118/119) is inferred by homology/annotation rather than demonstrated with mouse enzyme kinetics. Cross-species conservation of function is strong, but residue-level mouse data are limited.

2. **Physiological protease in vivo.** Chymosin is the classic in-vitro/industrial protease. Which protease(s) cleave mouse kappa-casein in the neonatal gut or mammary lumen physiologically, and whether a mouse GMP equivalent is released and bioactive in mouse pups, were not directly established in the reviewed literature.
3. **Chaperone and amyloid roles are largely in-vitro.** The holdase chaperone and amyloid-formation findings come from in-vitro and model-protein systems (often using bovine casein or human fragments). Their physiological significance specifically for mouse kappa-casein in vivo (e.g., preventing corpora amylacea in the mammary gland) is plausible but not directly demonstrated here.
4. **Quantitative structure.** The precise surface density, orientation, and dynamics of mouse kappa-casein on the micelle, and the exact stoichiometry with the calcium-sensitive caseins, remain inferred from bovine models.
5. **Glycosylation microheterogeneity.** Mucin-type O-glycosylation and sialylation of the macropeptide are heterogeneous and functionally consequential (GMP bioactivity is sialylation-dependent), but the mouse glycan structures and their functional variation were not characterized in detail.

Proposed Follow-up Experiments / Actions

1. **Mouse-specific cleavage mapping.** Perform in-vitro proteolysis of recombinant mouse kappa-casein with chymosin and with candidate neonatal gastric/intestinal proteases, followed by mass spectrometry, to confirm the res-118/119 scissile bond and identify the physiological protease in mouse pups.
2. **Structure–function of the macropeptide brush.** Generate knock-in mice bearing phospho-null (Ser/Thr→Ala) or glyco-null kappa-casein alleles to test whether the C-terminal PTMs are individually required for micelle stability and lactation, dissociating the structural from the bioactive-peptide functions.
3. **Cryo-EM / small-angle scattering of mouse micelles.** Apply SAXS/SANS contrast variation and cryo-EM directly to mouse milk to confirm that the bovine spongelike/nanocluster model holds, and to localize mouse kappa-casein at the surface.

4. **In-vivo chaperone test.** Examine *Csnk*$^{-/-}$ and macropeptide-mutant mammary glands for amyloid/corpora amylacea deposition to test whether kappa-casein's chaperone activity has a physiological proteostatic role in the lactating gland.
5. **Mouse GMP bioactivity.** Isolate the mouse kappa-casein macropeptide and test prebiotic/immunomodulatory activity (microbiota, TGF-β, mast cells) in mouse models to determine whether the bovine GMP bioactivities are conserved in the native species.
6. **Rescue experiment.** Test whether transgenic re-expression of wild-type vs. truncated (brush-less) kappa-casein rescues lactation in the *Csnk*$^{-/-}$ background, formally proving the macropeptide brush is the essential functional element.

Conclusion

Mouse *Csn3* encodes **kappa-casein**, a secreted, intrinsically disordered milk protein that functions as the essential colloidal stabilizer of the casein micelle. As the sole calcium-insensitive casein, it coats the calcium-phosphate/casein core and projects a phosphorylated, O-glycosylated/sialylated C-terminal macropeptide "brush" that sterically and electrostatically keeps milk calcium, phosphate, and the calcium-sensitive alpha-s/beta caseins in stable suspension. It acts extracellularly in secreted milk, with micelle assembly initiating in the Golgi of lactating mammary epithelial cells; its genetic deletion destabilizes micelles in the mammary lumen and abolishes lactation. It is also the defining physiological substrate of chymosin, whose cleavage of a single Phe–Met bond releases the bioactive glycomacropeptide and triggers milk coagulation.
